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Cis 130: Computer Organization

Assignment 3: Digital circuits, truth Tables, Boolean Algebra.

1) Create a truth table from the following boolean functions.

$$F1 = A'B'C + A'BC' + AB'C + ABC$$

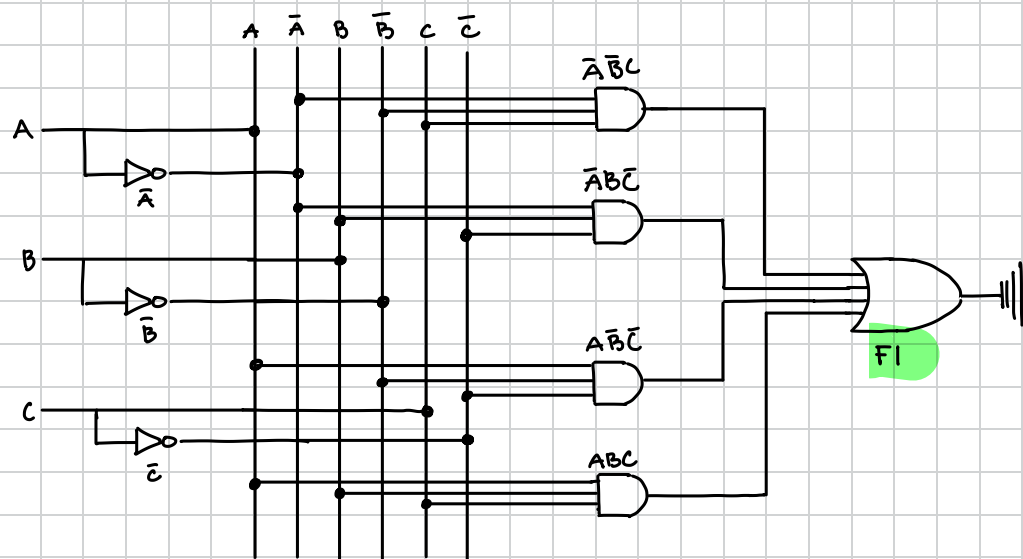
A	B	C	$\bar{A}$	$\bar{B}$	$\bar{C}$	$\bar{A}\bar{B}C$	$\bar{A}B\bar{C}$	$A\bar{B}C$	ABC	F1
0	0	0	1	1	1	0	0	0	0	0
0	0	1	1	1	0	1	0	0	0	1
0	1	0	1	0	1	0	1	0	0	1
0	1	1	1	0	0	0	0	0	0	0
1	0	0	0	1	1	0	0	0	0	0
1	0	1	0	1	0	0	0	1	0	1
1	1	0	0	0	1	0	0	0	0	0
1	1	1	0	0	0	0	0	0	1	1

$$F2 = AB + AC + BC$$

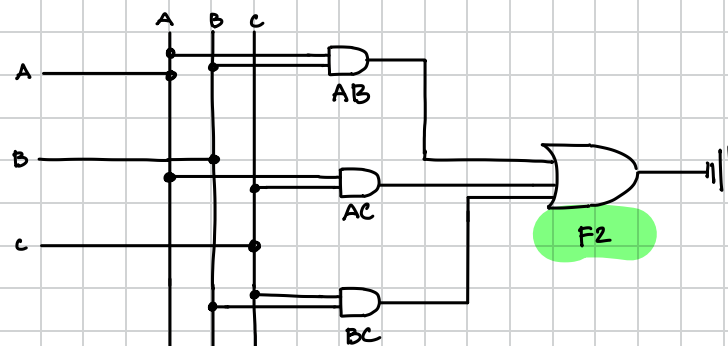
A	B	C	AB	AC	BC	F2
0	0	0	0	0	0	0
0	0	1	0	0	0	0
0	1	0	0	0	0	0
0	1	1	0	0	1	1
1	0	0	0	0	0	0
1	0	1	0	1	0	1
1	1	0	1	0	0	1
1	1	1	1	1	1	1

2) Draw the circuit diagrams derived from the following boolean functions:

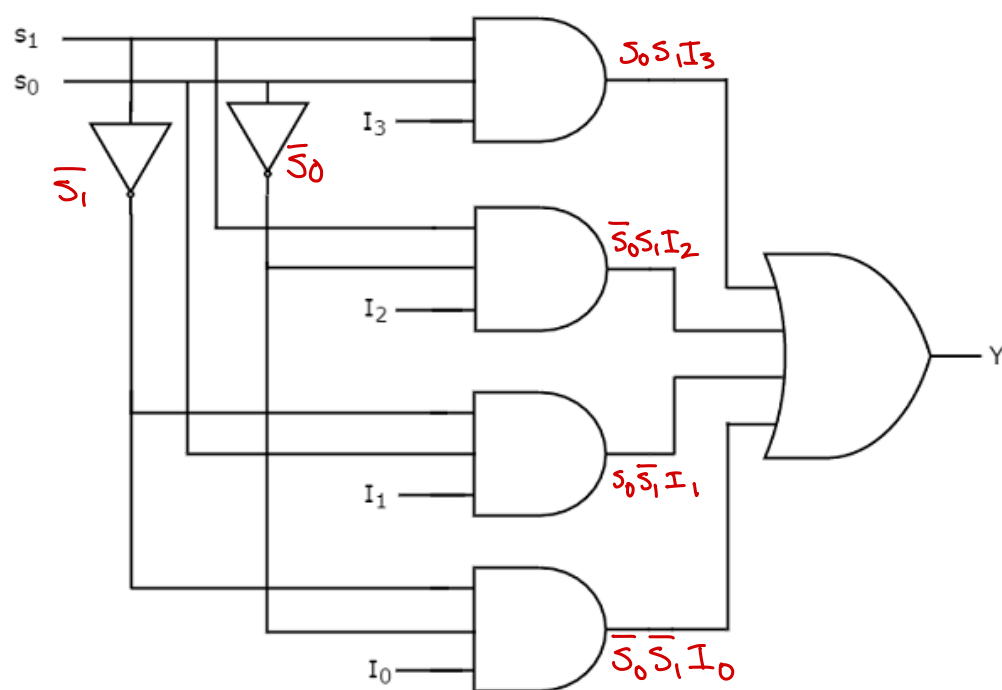
$$F1 = A'B'C + A'BC' + AB'C + ABC$$



$$F2 = AB + AC + BC$$



3) Write out a boolean function that describes the following circuit.



$$Y = (S_0 S_1 I_3) + (\bar{S}_0 S_1 I_2) + (S_0 \bar{S}_1 I_1) + (\bar{S}_0 \bar{S}_1 I_0)$$